1. Introduction

Dynamic Visual Cryptography for Secure Image Transmission addresses one of the most persistent problems in applied cryptography: how to move pictures across open networks without concentrating trust in a single key and without demanding heavy computation from the receiver. The system converts a secret image into a set of random-looking share images under a (k, n)-threshold rule. Any k or more shares reconstruct the secret by simple visual stacking or bitwise logic, while any smaller set reveals nothing useful, in an information-theoretic sense.

The qualifier **dynamic** distinguishes this work from classical visual cryptography. Classical schemes reuse fixed codebooks: the same secret pixel always maps to the same family of share patterns. Dynamic schemes inject fresh entropy in every session — a new seed, nonce, grid, or permutation — so that two transmissions of similar images produce statistically unlinkable shares. An adversary who collects shares across many sessions therefore gains no reusable dictionary.

Formally, let S be a secret image of size H x W with C channels. A (k, n) dynamic sharing produces n shares S1 ... Sn, each of size H x W, using a session key Ks drawn fresh per execution. Correctness requires that any subset of at least k shares recovers an approximation S' close to S under a defined visual metric, while secrecy requires that any subset of fewer than k shares is statistically independent of S. The session key Ks is never reused, and share pixels carry no positional markers that would aid alignment attacks.

Shares are first-class transmission objects. They can travel over disjoint network paths, rest on different cloud providers, or be held by distinct custodians. Loss, delay, or compromise of a minority of shares degrades availability gracefully rather than breaking secrecy catastrophically. This graceful degradation is a decisive advantage over single-ciphertext encryption, where loss of the key or corruption of the ciphertext is total.

Reconstruction is deliberately lightweight. Depending on the scheme variant, the receiver applies Boolean OR (physical stacking), XOR (digital recovery), or majority decoding per pixel location. No modular exponentiation, no S-box rounds, and no key schedule are needed at decode time. The design therefore suits browsers, low-end Android devices, bedside medical viewers, and microcontroller-adjacent edge gateways that cannot sustain AES-GCM at full frame rate.

The project scope spans the full pipeline rather than the sharing primitive alone. It covers image normalization, per-session random grid generation with cryptographically secure randomness, hash-tagged dispatch over simulated independent channels, strict integrity gating before any decode, quantitative scoring of both fidelity and secrecy, and an operator dashboard that makes threshold selection and forensic inspection accessible to non-cryptographers.

Representative deployments include inter-hospital transfer of radiology images, protection of fingerprint and iris templates in enrollment databases, distribution of survey and drone imagery among field teams, and escrow of board-level documents where no single administrator should hold the complete secret. In each case the security argument is uniform and auditable: below threshold, nothing leaks; at or above threshold with intact hashes, recovery succeeds.

Core Terminology

Term	Meaning in this project

Secret image	Original grayscale, color, or binary image to be protected
Share / shadow	One randomized output image; individually meaningless
(k, n) threshold	Any k of n shares recover; fewer than k reveal nothing
Dynamic session	Fresh randomness and grid per transmission; never reused
Random grid	Same-size stochastic encoding without pixel expansion
Integrity tag	SHA-256 hash per share used to reject tampering
Stacking recovery	OR / XOR / majority combine of authorized shares

System Components

1. **Secret Image Intake:** validates mode, dimensions, and channel layout before encoding.
2. **Dynamic Share Generator:** CSPRNG plus optional chaotic mixing produces n same-size shares.
3. **Transmission Module:** hash-tagged dispatch across independent logical channels.
4. **Verification Module:** recomputes hashes and enforces the k-of-n admission rule.
5. **Reconstruction Module:** stacking or XOR recovery with contrast handling.
6. **Evaluation Module:** PSNR, SSIM, entropy, correlation, and NPCR-style sensitivity.
7. **Dashboard:** single view of inputs, shares, channel status, output, and scores.

2. Background and Context of the Problem

Digital images now account for a dominant share of stored and transmitted bytes. A single hospital generates terabytes of X-ray, CT, MRI, and ultrasound frames per year; biometric programs hold millions of fingerprint and iris templates; mapping and agriculture pipelines stream high-resolution frames from drones; enterprises exchange scanned contracts and identity documents daily. Nearly all of this traffic crosses infrastructure the sender does not fully control: campus Wi-Fi, carrier networks, content delivery overlays, and multi-tenant object stores.

Classical image ciphers apply block or stream encryption to pixel bytes or compressed bitstreams. AES in CBC or GCM modes, and RSA-wrapped session keys, provide strong confidentiality when keys are managed well. The operational burden is nevertheless substantial: secure key generation, distribution, rotation, and revocation; per-pixel encryption and authentication on constrained senders; and full decryption before any viewing. A single leaked or escrowed key exposes every image protected under it, past and future, unless expensive re-encryption is performed.

Visual cryptography, founded by Naor and Shamir in 1994-1995, offered a qualitatively different trade-off. A binary secret pixel is expanded into m sub-pixels distributed across n transparencies according to complementary codebooks for white and black. Superposing any k transparencies reveals the secret to the naked eye; fewer reveal only random dots. No arithmetic is needed to decode. Subsequent work generalized the access structure, introduced grayscale and color handling via halftoning and channel splitting, and proposed meaningful (extended) shares that look like cover pictures rather than noise.

The price of classical constructions is well documented. Pixel expansion by a factor m = 2 to 4 multiplies storage and bandwidth. Contrast, defined as the relative difference between reconstructed white and black regions, drops to 1/m or worse, forcing larger fonts and thicker lines in practice. Most seriously for transmission security, the codebook is static: the same secret neighborhood always yields patterns from the same small family, so an observer accumulating many sessions can mount frequency and alignment analysis that would fail against a per-session randomized code.

Random grids, introduced by Kafri and Keren, removed expansion by encoding through stochastic grids of the same dimensions as the secret. Each share pixel is drawn from a controlled random process conditioned on the secret pixel and the threshold rule, rather than by expanding one pixel into many. Shares remain H x W, bandwidth matches the original, and contrast depends on statistical bias rather than geometric expansion. Early random-grid schemes were still static in deployment: one grid family reused indefinitely.

Dynamic visual cryptography closes that remaining gap by treating randomness as a per-session resource. A cryptographically secure pseudo-random number generator, optionally whitened by a chaotic map such as the logistic or tent map, is seeded with fresh operating-system entropy plus a session nonce and timestamp. The resulting keystream drives grid generation, pixel permutation, and share ordering. Because the keystream never repeats, share ensembles from different sessions are computationally indistinguishable even to an adversary who knows the algorithm and holds below-threshold shares from each session.

Network realities strengthen the case for threshold distribution. Packet loss, middlebox reordering, and regional outages routinely degrade individual paths. A (2, 3) or (3, 5) sharing lets the receiver recover as long as any authorized subset arrives intact, without retransmitting the entire image over a single fragile connection. Combined with per-share SHA-256 tags, the receiver can pinpoint which channel misbehaved and request only that share again, rather than restarting the whole transfer.

This project therefore sits at the intersection of four mature areas: threshold secret sharing, stochastic image encoding, hash-based transport integrity, and perceptual quality measurement. Its contribution is integrative and empirical: a complete, measurable, dashboard-operated transmission system in which dynamic secrecy, verified delivery, and quantified fidelity are demonstrated together rather than claimed separately.

Evolution of Visual Cryptography

Era	Development	Effect
1987	Random grids (Kafri-Keren)	Same-size encoding without expansion
1994-1995	Threshold VC (Naor-Shamir)	Formal k-of-n visual sharing

1996-2003	General access, color VC	Flexible parties; grayscale and color support
2004-2011	Probabilistic, RG-threshold	Better contrast; formal RG threshold proofs
2012-2020	Chaotic, meaningful shares	Per-session entropy; usable cover shares
2020-2026	Dynamic transmission systems	Verified delivery plus joint quality scoring

3. Objectives of the Project

The central objective is to specify, implement, and validate a dynamic threshold image transmission system that is secure under repeated observation, economical in bandwidth, verifiable in transit, and operable through a single dashboard. Seven specific objective groups decompose that goal into testable deliverables, each with acceptance criteria used later in evaluation.

1. Image Ingestion and Normalization

- Accept binary, grayscale, and 24-bit color images up to 2048 x 2048 without manual conversion.
- Normalize mode, orientation metadata, and alpha handling deterministically.
- Split color inputs into independent channels that share one session nonce.

Acceptance: a mixed corpus of at least 60 images across modes and sizes loads without error and round-trips through a null share-decode path with PSNR above 40 dB before any secrecy transform is applied.

2. Per-Session Dynamic Share Generation

- Implement (2, 2), (2, 3), (3, 5), and generic (k, n) random-grid encoders.
- Guarantee same-size shares: no pixel expansion under any setting.
- Bind every share set to a fresh 256-bit session seed logged with timestamp.

Acceptance: re-encoding the same image twice yields share sets with cross-correlation below 0.05 and NPCR above 99 percent, confirming session unlinkability.

3. Randomness Quality and Secrecy

- Source entropy from OS CSPRNG; optionally mix logistic-map output.
- Test single-share entropy above 7.9 bits per pixel on 8-bit channels.
- Test below-threshold share-to-secret correlation within plus-minus 0.05 of zero.

4. Integrity-Protected Transmission

- Emit SHA-256 per share plus a signed manifest of (k, n), dimensions, and nonce.
- Simulate independent channels with configurable loss, delay, and tamper injection.
- Support selective retransmission of only failed shares.

5. Gated Reconstruction

- Admit only hash-valid shares; abort with diagnostics when fewer than k are valid.
- Offer OR stacking for visual inspection and XOR for digital fidelity.
- Apply contrast normalization and channel reassembly for color outputs.

6. Joint Quality and Security Scoring

- Report PSNR, SSIM, and Michelson contrast retention for every reconstruction.
- Report entropy, correlation, and differential sensitivity for every share set.
- Persist scores per session for threshold comparison and regression testing.

7. Integration, Usability, and Reproducibility

- Deliver one command-to-dashboard path from image selection to scored output.
- Log seeds, parameters, hashes, and scores for exact session replay.
- Validate on benchmark, medical-sample, biometric-sample, and self-collected images.

4. Problem Statement and Definition

Problem Statement

Mission-critical images are routinely sent over paths where passive eavesdropping and active tampering are both realistic. Conventional encryption answers with a single data key: whoever holds it reads everything. Key distribution to multiple receivers, timely revocation after staff changes, and decryption on underpowered viewers each add operational friction. When the key is mishandled, the failure is total and often silent until disclosed externally.

Threshold visual cryptography would appear to solve the single-key fragility by splitting trust. In practice, deployed variants retain three structural weaknesses. First, static codebooks and reused grids let a persistent observer correlate shares across sessions and progressively narrow the secret distribution. Second, pixel expansion inflates every share, penalizing precisely the bandwidth-constrained links where lightweight protection matters most. Third, shares travel without intrinsic integrity: a flipped bit, a cropped border, or a substituted share produces a plausible-looking but wrong reconstruction that the receiver cannot distinguish from the truth.

The operational question is therefore not whether visual sharing works in principle — it does — but whether a complete transmission system can deliver per-session unlinkability, same-size shares, tamper-evident delivery, flexible thresholds, and measured fidelity together, with an interface that working engineers and students can operate and audit. No turnkey baseline combining all five properties is available for classroom and laboratory use.

Threat and Failure Model

- Passive observer: captures any single channel long-term; must learn nothing from below-threshold shares.
- Active tamperer: flips pixels, crops, or replays old shares; must be detected by hash gating.

- Lossy network: drops or delays whole shares; system must recover from any authorized subset.
- Insider minority: fewer than k custodians collude; must remain unable to reconstruct.
- Replay adversary: resubmits a prior valid share into a new session; must fail nonce binding.

Specific Deficiencies Addressed

1. **Session linkability** — static families of patterns leak across repeated encryptions.
2. **Expansion overhead** — m-fold larger shares waste storage and airtime.
3. **Silent corruption** — no integrity tags means wrong images look right.
4. **Rigid participation** — fixed (k, n) cannot track changing teams or paths.
5. **Unmeasured trade-offs** — contrast, size, and secrecy are tuned blind without joint metrics.

Problem Definition

Design, build, and measure a **Dynamic Visual Cryptography for Secure Image Transmission** system in which each transmission draws fresh entropy, emits n same-size hash-tagged shares under a configurable (k, n) rule, admits only verifiably intact shares, reconstructs from any authorized subset by lightweight logic, and reports standardized fidelity and secrecy scores. The definition of done is operational: unlinkable sessions, verified delivery, graceful degradation under loss, and reproducible numbers guiding threshold choice.

Out of scope for the core guarantee are endpoint compromise beyond threshold, denial-of-service against all channels simultaneously, and physical capture of an authorized quorum of displays. Those risks are acknowledged and handled by surrounding operational controls, not by the sharing primitive itself.

5. Need and Significance of the Study

Need for the Study

- Constrained receivers need decode without big-number arithmetic or round-heavy ciphers.
- Repeated transmissions need fresh share statistics to defeat longitudinal analysis.
- Regulated images need tamper evidence, not just secrecy, to support clinical and legal reliance.
- Distributed operations need quorum recovery when individual links or stores fail.
- Teaching laboratories need a runnable, inspectable baseline instead of isolated equations.
- Procurement needs numbers: PSNR, SSIM, entropy, and correlation per threshold setting.

Technical Significance

The study provides controlled evidence on three contested points: whether per-session renewal measurably reduces cross-session correlation at fixed contrast; whether random-grid threshold sharing sustains usable SSIM without expansion across image classes; and whether hash gating can be added without changing the secrecy argument. Each claim is paired with a procedure and a threshold for acceptance, moving discussion from assertion to measurement.

- Quantifies unlinkability via cross-session NPCR and correlation tests.

- Quantifies fidelity cost of threshold choice on a shared corpus.
- Isolates integrity overhead in bytes and milliseconds per share.

Security and Operational Significance

Operationally, the system replaces one high-value key with n low-value shares. Compromise of a single file server, mailbox, or handset yields noise. Rotation means re-sharing under a new nonce rather than re-keying an installed base. Forensics improve because each share carries its own hash and session binding: investigators can state exactly which channel failed and when.

- Blast radius limited to below-threshold holdings.
- Selective retransmission instead of full re-encryption.
- Auditable manifest per transmission for compliance review.

Economic Significance

Bandwidth and compute dominate image-system operating cost at scale. Same-size shares avoid the 2x to 4x expansion tax of classical VC. Lightweight decode avoids provisioning cryptographic accelerators on every viewer. Early rejection of corrupt shares avoids wasted radiologist or analyst time spent interpreting a subtly wrong image. Together these savings compound across archives holding millions of frames.

Societal Significance

Privacy-preserving handling of diagnostic and biometric images directly affects patient and citizen trust. A demonstrably verifiable sharing pipeline supports tele-radiology across low-bandwidth rural links, second-opinion workflows without surrendering full archives to any single intermediary, and enrollment systems where template custodians cannot individually abuse their holdings. The societal return iscнosafer delegation of sensitive imagery to necessary parties and no one else.

6. Existing System / Existing Work

1. Conventional Cipher-Based Image Protection

Production systems most often encrypt raster bytes with AES-128/256 in CBC, CTR, or GCM modes, wrapping the data key with RSA or ECDH for distribution. Authenticated modes add integrity at the cost of nonces and tags per message. Format-preserving variants and selective encryption of DCT coefficients reduce cost for JPEG and H.264 payloads but couple security to codec internals.

Strengths are standardization, hardware acceleration, and well-understood proofs. Weaknesses for this use case are key-management weight, all-or-nothing failure, and decode cost on weak viewers. Loss of one share in the proposed system is an inconvenience; loss of one key in a cipher system is a breach or an outage.

2. Classical Naor–Shamir Threshold Sharing

The original (2, 2), (2, n), and general (k, n) constructions expand each secret pixel into m sub-pixels using basis matrices, guaranteeing perfect secrecy below threshold and deterministic stacking recovery above it. Contrast is 1/m in the basic case, improvable with careful matrix design at the cost of larger m.

The construction is elegant and provably secure, but static codebooks, expansion, and contrast loss limit transmission appeal. Every deployment using the same matrices shares the same pattern family, which is precisely the longitudinal signal a dynamic system seeks to eliminate.

3. Extended, Friendly, and Progressive Variants

Extended VC replaces noise-like shares with meaningful cover images by embedding the secret in halftone textures. Progressive VC layers multiple secrets or quality tiers so that stacking more shares yields progressively clearer output. Friendly shares ease handling and reduce suspicion during transport.

Usability improves, yet the underlying encoding typically remains session-invariant. Covers can even help an attacker by providing known-texture regions for differential comparison across sessions. Dynamic renewal is orthogonal and can be layered onto these variants.

4. Random-Grid Threshold Schemes

Kafri-Keren grids and their threshold descendants (Shyu, Chen-Tsao, Yang) encode by per-pixel coin tosses conditioned on the secret and the access rule. No expansion occurs; contrast emerges from bias between black and white reconstruction distributions. Proofs establish that below-threshold grids are uniformly random and independent of the secret.

These schemes supply the encoding core adopted here. Their limitation in prior deployments is operational rather than mathematical: grids drawn from a long-lived stream without explicit session binding and without transport integrity. This project retains the mathematics and hardens the surrounding system.

5. Chaotic and CSPRNG-Driven Randomness Layers

Logistic, tent, sine, and Lorenz-derived sequences, often post-processed by cryptographic hashing, have been proposed as lightweight entropy sources for image ciphers and share generators. Statistical test suites (NIST SP 800-22, TestU01) are commonly reported. When seeded correctly and reseeded per session, chaotic layers add unpredictability and break cross-session alignment.

The risk is undisciplined use: fixed chaotic parameters and seeds reproduce the static-codebook flaw in new clothing. The project therefore mandates OS-seeded CSPRNG output as the authoritative entropy, with chaotic mixing strictly auxiliary and always session-bound.

6. Hybrid VC plus Conventional Encryption

Some pipelines encrypt shares with AES or embed them steganographically in carriers for defense in depth. Confidentiality then rests on both layers, at the price of restoring heavy decode and key management at every receiver.

Hybrids are justified where regulatory baselines mandate approved ciphers on the wire. The proposed architecture accommodates an optional transport cipher without making it load-bearing for the secrecy proof: even with the outer layer stripped, below-threshold shares remain uninformative.

7. Steganography and Watermarking Approaches

Least-significant-bit embedding, transform-domain hiding, and robust watermarking conceal a payload inside a cover image. Detection resistance and robustness to compression are the primary metrics. Capacity, imperceptibility, and survivability trade against one another.

Hiding differs fundamentally from sharing: possession of the stego image plus the extraction key yields the whole secret, whereas possession of a minority of shares yields nothing regardless of effort. Hiding complements sharing for cover transport but cannot substitute for threshold trust-splitting.

8. Cloud, IoT, and Telemedicine Transmission Frameworks

Recent applied work distributes medical and sensor imagery across multi-cloud buckets, MQTT and CoAP relays, and edge gateways with store-and-forward behavior. Emphasis is on availability, latency budgets, and audit trails. Security is commonly delegated entirely to TLS and at-rest encryption, reintroducing single-credential fragility at the broker or bucket.

Threshold sharing integrates naturally with these topologies: each broker or bucket holds at most a minority of shares, so a compromised relay learns nothing while availability benefits from path diversity. Hash gating supplies the per-hop integrity that bare pub-sub lacks.

Comparative Summary

Approach	Core mechanism	Key limitation for this use
AES / RSA imaging	Block / public-key cipher	Single-key failure; heavy decode
Naor-Shamir VC	Basis matrices	Static; expansion; low contrast
Extended / progressive	Cover / layered codebooks	Usable but still static
Random-grid threshold	Stochastic same-size grids	Often static deployment; no transit integrity
Chaotic-only ciphers	Deterministic chaos stream	Seed reuse risk; uneven analysis
Hybrid VC+cipher	VC plus AES layer	Restores heavy decode everywhere
Steganography	Cover embedding	No threshold guarantee
Cloud / IoT TLS-only	Transport encryption	Broker credential is single point of trust

7. Research Gap / Limitations of Existing Work

Surveyed literature establishes each ingredient in isolation. What remains scarce is a single reproducible system that demonstrates dynamic renewal, verified transport, flexible thresholds, and joint scoring together on a shared corpus with stated acceptance limits. Eight concrete gaps shape the present design.

1. **Session-bound evaluation is thin:** most secrecy reports analyze one share set, not pairs of sets from repeated encryptions of related images. Cross-session NPCR and correlation are seldom published, leaving unlinkability unquantified.
2. **Integrity is an afterthought:** hash tagging, nonce binding, and admission policy are described qualitatively if at all, without byte overhead or detection-rate measurement under injected tampering.
3. **Color handling is uneven:** many threshold proofs and experiments cover binary images; channel-independent versus vector color rules and their SSIM consequences are under-compared.

4. **Threshold operating curves are missing:** PSNR, SSIM, and contrast as functions of (k, n) on one corpus are rarely plotted together, forcing blind parameter choice in deployment.
5. **Randomness discipline varies:** chaotic parameters, seeding, and reseeding practice differ across papers, hindering fair comparison and inviting accidental determinism.
6. **Loss and tamper robustness is simulated narrowly:** burst loss of whole shares, mixed loss-plus-tamper, and replay of stale shares are seldom combined in one harness.
7. **Usability and replayability lag:** few artifacts let an examiner vary (k, n), inspect shares, and replay a session bit-for-bit from logged seeds and manifests.
8. **Longitudinal adversary models are informal:** persistent multi-session observers with below-threshold holdings from each session lack a standard game definition in applied VC papers.

Project-level note: no claim is made that dynamic grids, chaotic mixing, or threshold sharing are individually novel. Each has peer-reviewed precedent. The gap — and the contribution — is integrative: binding them into one verified, measured, operable transmission baseline with explicit acceptance criteria and a publicizable evaluation procedure.

8. Aim of the Project

Aim 1 — Dynamic secrecy with same-size shares. Build a (k, n) random-grid sharing engine in which every transmission draws a fresh 256-bit session seed, emits n shares of identical dimensions to the secret, and guarantees measured unlinkability: cross-session share correlation within plus-minus 0.05 of zero and NPCR above 99 percent on re-encryption of identical inputs.

Aim 2 — Verified, quorum-gated delivery. Deliver each share with a SHA-256 tag and a session-bound manifest; admit a share to reconstruction only on hash match and nonce match; proceed only when at least k valid shares are present; otherwise abort with a channel-level diagnostic. Target zero undetected tamper admissions across injected single-pixel, crop, and substitution suites.

Aim 3 — Measured fidelity and operability. Reconstruct by OR stacking and XOR logic with channel-aware reassembly, score every session with PSNR, SSIM, contrast retention, entropy, and correlation, and expose the full loop through a dashboard that supports threshold comparison, share inspection, and bit-exact replay from logged parameters. Validate on benchmark, medical-sample, biometric-sample, and self-collected corpora.

9. Literature Survey / Review of Related Work

Fifteen representative works trace the path from fixed codebooks to session-renewed transmission systems. Each entry summarizes the contribution, states the limitation most relevant to transmission, and notes the specific design response adopted in this project.

Study 1 — Naor and Shamir (1994-1995): Threshold visual cryptography

Contribution: formal (k, n) sharing with basis matrices; perfect secrecy below threshold; recovery by stacking without computation. Established contrast and expansion as primary design parameters.

Limitation and response: static codebooks with expansion m >= 2. Project response: retain threshold semantics but replace fixed matrices with per-session random grids to remove expansion and session linkability.

Study 2 — Kafri and Keren (1987): Random grids

Contribution: stochastic same-size encoding; shares indistinguishable from noise; no codebook distribution problem. Showed that randomness alone can carry the secrecy burden.

Limitation and response: binary focus without threshold formalization or transport handling. Response: adopt grids as encoder core inside a full (k, n) threshold and integrity framework.

Study 3 — Ateniese et al. (1996): General access structures

Contribution: extended threshold logic to arbitrary qualified and forbidden sets, enablingNeed-to-know coalitions beyond simple k-of-n. Important for organizational custody rules.

Limitation and response: constructions remain static and expansion-prone. Response: implement the common (k, n) special case dynamically; structure the manifest so general policies can be added later.

Study 4 — Hou (2003) and color VC line: Color extensions

Contribution: halftone, color-decomposition, and vector rules carrying VC to grayscale and color with MK-style contrast analysis. Made photographic and medical images addressable.

Limitation and response: contrast penalty and channel-correlation leakage. Response: channel-independent encoding with per-channel scoring, plus explicit SSIM reporting per image class.

Study 5 — Yang (2004): Probabilistic sharing

Contribution: probabilistic method reducing expansion at the cost of variance; showed expected-contrast analysis and variance-aware parameter choice.

Limitation and response: single-session analysis. Response: repeat probabilistic draws under fresh session seeds and report variance across sessions, not just within one.

Study 6 — Shyu (2007): Image encryption by random grids

Contribution: rigorous treatment of grid-based encryption for binary and grayscale images; clarified randomness requirements and visual quality behavior.

Limitation and response: deployment treated as one-shot encryption. Response: wrap grid primitives in session management, manifests, and retransmission logic.

Study 7 — Wang et al. (2007): Boolean-operation sharing

Contribution: two sharing constructions based purely on Boolean operations, simplifying implementation and enabling XOR-based lossless paths alongside OR stacking.

Limitation and response: static Boolean rules. Response: offer both OR and XOR recovery modes under dynamic keying so operators can trade visual inspection against digital fidelity.

Study 8 — Chen and Tsao (2009-2011): Threshold random grids

Contribution: formal (k, n) random-grid threshold with secrecy proofs and contrast characterization; demonstrated practical same-size multi-share generation.

Limitation and response: limited transport and integrity treatment. Response: add hash gating, nonce binding, and loss-aware admission as first-class pipeline stages.

Study 9 — Chaotic-map image ciphers (2010-2018 wave)

Contribution: logistic, tent, and Lorenz-driven keystreams with histogram, entropy, NPCR, and UACI reporting; fast software generation and large key spaces claimed.

Limitation and response: inconsistent seeding discipline and overclaimed key spaces. Response: use OS CSPRNG as authoritative entropy; treat chaotic output as auxiliary whitening only, always reseeded per session.

Study 10 — Extended and meaningful-share VC (2010-2019)

Contribution: cover-carrying shares that avoid attracting attention; techniques for balancing cover fidelity against secret contrast.

Limitation and response: covers create known-plaintext footholds for cross-session differencing. Response: default to noise shares for maximum secrecy; reserve meaningful shares as an explicitly marked option.

Study 11 — Progressive and multi-secret VC (2012-2020)

Contribution: stacked quality tiers and multiple secrets per share set, improving bandwidth efficiency for layered disclosure workflows.

Limitation and response: complex decoding order and heightened sensitivity to share loss. Response: defer multi-secret support; guarantee single-secret quorum robustness first, with manifest versioning for future tiers.

Study 12 — Visual VC for medical and biometric handling (2015-2023)

Contribution: domain pilots showing VC for X-ray, fundus, fingerprint, and iris protection with clinician- and matcher-centered quality checks.

Limitation and response: small private corpora and ad hoc metrics. Response: evaluate on a mixed public-plus-sample corpus with uniform PSNR, SSIM, entropy, and correlation reporting for comparability.

Study 13 — IoT and edge secure imaging (2019-2024)

Contribution: lightweight ciphers and selective-encryption schemes for MQTT, CoAP, and gateway topologies under latency and energy budgets.

Limitation and response: reliance on broker credentials as trust roots. Response: map threshold shares onto distinct brokers or topics so no single relay holds a quorum.

Study 14 — Hash-chained and blockchain-anchored sharing (2020-2025)

Contribution: manifests anchored in tamper-evident logs for auditability; smart-contract-mediated quorum release.

Limitation and response: ledger overhead and availability coupling. Response: keep verification local and offline-capable via SHA-256 manifests; design manifest format to permit optional anchoring later.

Study 15 — Quality-and-security joint benchmarks (2021-2026)

Contribution: emerging practice of reporting PSNR, SSIM, entropy, correlation, NPCR, and UACI together with timing and size overhead on shared image sets.

Limitation and response: procedures still vary. Response: freeze one joint scorecard in the dashboard and apply it unchanged across all (k, n) settings and image classes in this project.

10. Proposed Methodology

Methodology is organized as seven sequential phases with explicit inputs, transforms, and gates. A session either passes every gate with logged evidence or aborts with a diagnostic identifying the failed phase. All randomness is drawn per session; no state crosses session boundaries except the versioned algorithm identifier.

Phase 1 — Acquisition and Normalization

Input: file or capture buffer plus operator-selected (k, n) and recovery mode (OR or XOR). Processing: detect mode (1-bit, 8-bit gray, 24-bit color), strip or record orientation metadata, pad or tile to an even lattice only if the threshold construction requires it, and split color into R, G, B planes sharing one session nonce. Output: canonical H x W x C array, manifest skeleton, and a null-path fidelity check exceeding 40 dB PSNR.

Phase 2 — Session Entropy and Grid Derivation

Draw 256 bits from the OS CSPRNG, concatenate operator nonce, timestamp, and image digest, and expand via HKDF-SHA256 into per-channel keystreams. Optionally xor-mix a logistic-map stream ($r = 3.99$, freshly seeded per session) as auxiliary whitening. Derive one H x W random grid per required share from the keystream using a documented indexing order. Rationale: fresh keystreams give unlinkability; HKDF binding gives replay resistance; auxiliary chaos adds diffusion without becoming load-bearing.

Phase 3 — Threshold Encoding (No Expansion)

Apply the (k, n) random-grid rule per pixel location: for white secret pixels emit share bits with one biased distribution, for black with the complementary distribution, such that any k shares show the correct bias while any smaller set is uniform. Process channels independently under the same session key. Complexity is O(n x H x W) time and O(n x H x W) bytes resident only during dispatch; streaming row-bands bound memory to O(n x W). Record per-share digests immediately.

Phase 4 — Tagging and Dispatch

Compute SHA-256 over each share's canonical bytes, assemble a manifest containing algorithm version, (k, n), dimensions, mode, session identifier, timestamps, and the tag list, and dispatch shares over logically independent channels (separate sockets, directories, or object prefixes). Simulate adverse conditions via injectors for drop, delay, single-pixel flip, crop, and stale-share replay. Log per-channel acknowledgement or timeout.

Phase 5 — Admission Gating and Verification

On receipt, recompute every tag, check manifest signature or checksum, verify nonce freshness within the allowed window, and enforce the quorum rule: proceed only if at least k shares are both present and valid. Failures produce specific codes (MISSING, TAMPER, STALE, PARAM_MISMATCH) rather than a best-effort decode. This strictness is intentional: a wrong image is worse than no image in clinical and legal contexts.

Phase 6 — Reconstruction and Post-Processing

Combine admitted shares by pixel-wise OR for visual stacking or XOR for digital recovery per operator mode. Reassemble color channels, apply range clipping and optional unsharp masking for OR outputs, and preserve raw XOR outputs bit-exact for archival comparison. Record decode latency and output digests alongside input digests for end-to-end traceability.

Phase 7 — Scoring, Comparison, and Replay

Score fidelity with PSNR = 10 log10(MAX^2 / MSE), SSIM over luminance-contrast-structure windows, and Michelson contrast retention (Cout / Cin). Score secrecy with per-share Shannon entropy (target above 7.9 on 8-bit data), share-to-secret Pearson correlation (target within plus-minus 0.05 of zero), and cross-session NPCR above 99 percent on re-encryption. Persist parameters, seeds (or seed commitments), hashes, timings, and scores per session; any session replays bit-exact from its log.

Complexity and Overhead Summary

Cost dimension	Behavior
Encode time	O(n x H x W); linear in pixels and shares
Decode time	O(k x H x W); linear in quorum size
Share size	H x W each; zero expansion
Integrity bytes	32 bytes SHA-256 per share plus small manifest
Memory	Row-streamed; O(n x W) with banding
Latency driver	Slowest admitted channel; selective retry only

11. Proposed System / Architecture / Framework

The system realizes the methodology as six loosely coupled layers communicating through versioned data contracts (canonical image, share envelope, manifest, scorecard). Layers can be tested in isolation with golden vectors and composed into the full pipeline without hidden shared state. The dashboard orchestrates layers and renders every intermediate artifact for inspection.

Working Flow

Secret Image → Dynamic Share Generator → Secure Transmission → Integrity Verification → Share Reconstruction → Reconstructed Image + Scorecard Dashboard

Control flow is gate-driven: normalization must certify the canonical array; generation must emit tags; dispatch must log per-channel receipts; verification must admit a quorum; reconstruction runs only on admitted inputs; scoring runs only on certified outputs. Any gate failure halts the session with a machine-readable code and a human-readable explanation, preventing silent wrong-image delivery.

1. Preprocessing Layer

Responsibilities: format detection, mode normalization, metadata policy, channel split, manifest skeleton. Interfaces: accepts common raster inputs; emits canonical arrays or typed errors (UNSUPPORTED_MODE, DIMENSION_LIMIT). Testability: null-path round-trip fixtures with PSNR assertions.

2. Dynamic Share Generation Layer

Responsibilities: entropy intake, keystream expansion, grid derivation, threshold encoding, digest emission. Interfaces: consumes canonical array plus (k, n, mode, seed); produces n share envelopes. Testability: deterministic fixtures under fixed seeds plus statistical suites under fresh seeds.

3. Secure Transmission Layer

Responsibilities: channel abstraction, tag attachment, retry policy, adverse-condition injection for evaluation. Interfaces: share envelopes in, per-channel receipts out. Testability: scripted loss, delay, flip, crop, and replay scenarios with expected admission outcomes.

4. Integrity Verification Layer

Responsibilities: tag recomputation, nonce and manifest checks, quorum enforcement, diagnostic coding. Interfaces: received envelopes plus manifest in, admitted quorum or abort code out. Testability: mutation suites where exactly one corruption class is active per case.

5. Reconstruction Layer

Responsibilities: OR / XOR combine, channel reassembly, range handling, latency accounting. Interfaces: admitted quorum plus mode in, reconstructed image plus digests out. Testability: golden share sets with known outputs per (k, n, mode).

6. Evaluation and Monitoring Layer

Responsibilities: fidelity scoring, secrecy scoring, cross-session comparison, persistence, dashboard rendering. Interfaces: inputs, shares, outputs, and logs in; scorecard and plots out. Testability: reference images with precomputed PSNR and SSIM anchors.

Failure Handling

Failure	Detection	Response
Share lost in transit	Timeout without receipt	Recover if quorum remains; else selective retry
Pixel flip / crop	SHA-256 mismatch	Quarantine share; continue if quorum remains
Stale replay	Nonce / timestamp check	Reject with STALE code
Below-quorum arrival	Count of valid shares < k	Abort with MISSING diagnostic
Parameter mismatch	Manifest comparison	Abort with PARAM_MISMATCH

12. Tools, Technologies and Datasets

Tooling is chosen for auditability and laboratory availability: mainstream Python imaging and numeric libraries, standard hash primitives, and a browser dashboard with no proprietary dependencies. Versions are pinned in requirements and recorded per session so results remain reproducible across machines.

1. Software Stack

- **Python 3.10+:** orchestration, encoding logic, and test harnesses.
- **NumPy 1.24+:** vectorized pixel and grid operations.
- **OpenCV 4.x / Pillow 10.x:** decoding, normalization, and rendering.
- **hashlib (SHA-256) / PyCryptodome:** tags and optional transport wrapping.
- **Matplotlib:** share montages and metric plots.
- **Streamlit:** operator dashboard for thresholds, status, and scores.
- **Jupyter / Colab:** interactive analysis and demonstration notebooks.

2. Security Primitives

- **OS CSPRNG (secrets / os.urandom):** authoritative per-session entropy.
- **HKDF-SHA256:** keystream expansion bound to session context.
- **Logistic / tent auxiliary stream:** non-load-bearing diffusion layer.
- **SHA-256 manifests:** per-share integrity plus session binding.

3. Transmission Harness

- **TCP sockets / directory drops:** independent-channel abstraction.

- **Fault injector:** drop, delay, flip, crop, replay with seeded scripts.
- **Logger:** JSONL per session: parameters, hashes, timings, scores.

4. Requirements and Datasets

Item	Baseline used
CPU / RAM	Quad-core, 8 GB; no GPU required
OS	Windows 10/11 or Ubuntu 22.04
Benchmark images	USC-SIPI: Lena, Peppers, Mandrill, House (gray + color)
Medical samples	Anonymized X-ray / MRI teaching samples, de-identified
Biometric samples	Public-domain fingerprint / iris teaching samples
Self-collected set	60+ images across sizes, lighting, and textures

Dataset ethics: only openly licensed or laboratory-captured images are used. Any human-derived samples are de-identified teaching images; no live biometric enrollment data is handled. The self-collected set documents capture device, illumination, and resolution so that generality claims can be inspected rather than assumed.

13. Expected Outcomes

1. **Session unlinkability demonstrated:** re-encryption NPCR above 99 percent with negligible cross-correlation.
2. **Zero-expansion sharing:** every share exactly H x W across all tested thresholds.
3. **Tamper-evident delivery:** 100 percent rejection of single-pixel, crop, and substitution corruptions in fault suites.
4. **Quorum-robust recovery:** any k valid shares reconstruct; any smaller set fails cleanly.
5. **Dual-scored sessions:** every reconstruction carries PSNR, SSIM, entropy, and correlation.
6. **Threshold operating curves:** fidelity versus (k, n) plotted on one corpus for deployment choice.
7. **Selective repair:** only failed shares retransmitted after diagnosed faults.
8. **Replayable evidence:** bit-exact session replay from persisted logs.
9. **Operable dashboard:** non-expert walkthrough from image to scored output under ten minutes.
10. **Extensible baseline:** versioned contracts admit video-frame and multi-cloud extensions.

Quantitative Targets

Metric	Acceptance target
PSNR (XOR path)	Above 30 dB on benchmark set
SSIM (XOR path)	Above 0.90 on benchmark set

Share entropy (8-bit)	Above 7.9 bits per pixel
Share-to-secret correlation	Within +-0.05 of zero
Cross-session NPCR	Above 99%
Tamper detection rate	100% on injected suites
Below-quorum leakage	No visual or statistical recovery

14. Scope of the Project

In Scope

1. **Multi-mode images** — binary, grayscale, and 24-bit color intake and scoring.
2. **Dynamic (k, n) sharing** — (2, 2), (2, 3), (3, 5), and generic thresholds.
3. **Session management** — fresh entropy, nonces, and manifests per transmission.
4. **Simulated multi-channel transport** — loss, delay, and tamper injection.
5. **Strict integrity gating** — hash and quorum enforcement before decode.
6. **Dual recovery modes** — OR stacking and XOR digital paths.
7. **Joint scorecard** — fidelity plus secrecy plus timing per session.
8. **Dashboard operation** — selection, inspection, comparison, and replay.

Out of Scope (with path forward)

1. Live wide-area deployment across carriers (simulated channels stand in).
2. Multi-secret and progressive-tier encoding (manifest versioned for later).
3. Video-rate streaming optimization (frame-wise sharing is the natural extension).
4. Ledger-anchored audit (manifest format permits optional anchoring).
5. Endpoint hardening and access control (assumed from host environment).

Overall scope: a complete laboratory transmission baseline that proves dynamic threshold sharing can be delivered verifiably and measured honestly, ready for extension to live networks, video, and cloud-native quorum storage.

15. Limitations and Assumptions

Limitations and Mitigations

1. **OR-path contrast loss:** stacked viewing is dimmer than XOR digital output. Mitigation: dual-mode recovery with guidance on when each applies.
2. **Per-session encode cost:** fresh entropy and grid derivation add sender work linear in n. Mitigation: vectorized NumPy paths and row banding.

3. **Quorum availability:** fewer than k intact shares cannot recover by design. Mitigation: choose (k, n) to match measured channel reliability plus selective retry.
4. **Simulated network:** real carrier behavior may differ. Mitigation: parameterize loss and delay from traces where available; keep channel abstraction narrow.
5. **Corpus limits:** benchmark plus samples cannot cover all imagery. Mitigation: publish corpus manifest and per-image scores for external replication.
6. **Auxiliary chaos fragility:** poorly chosen maps degrade rather than help. Mitigation: chaos strictly auxiliary; CSPRNG output authoritative.
7. **No DoS resistance:** suppression of all channels defeats availability. Mitigation: path diversity and monitoring, outside the sharing primitive.
8. **Viewer calibration:** perceived contrast varies by display and print. Mitigation: report instrumented SSIM and contrast alongside visual inspection.

Assumptions

1. Inputs conform to standard raster modes processable by OpenCV or Pillow.
2. Operating-system entropy is available and uncompromised on the sender.
3. At least k shares can be delivered intact within the retry budget.
4. Parties agree on algorithm version, (k, n), and recovery mode before the session.
5. SHA-256 remains second-preimage resistant for the deployment lifetime.
6. PSNR, SSIM, entropy, and correlation are accepted proxies for task suitability, with domain review before clinical or legal reliance.

16. Expected Applications

1. **Inter-hospital radiology transfer:** route shares over separate VPNs or media; either hospital alone holds nothing diagnostic until quorum assembly under audit.
2. **Biometric template custody:** split enrollment templates so no single database row or administrator can impersonate; matching executes only on quorum join in a hardened enclave.
3. **Defence and field imagery:** distribute drone or survey frames across couriers and links; capture of one courier yields noise while quorum recovery tolerates single-link outage.
4. **Multi-cloud archival:** place shares in distinct providers or regions; provider breach or billing lockout degrades to reduced redundancy, not disclosure.
5. **Board and legal document escrow:** require k-of-n officers to assemble scanned resolutions; below-quorum holdings satisfy least-privilege review.
6. **IoT visual sensing:** publish shares to distinct MQTT topics or gateways; constrained cameras perform only lightweight encoding while verification runs at the aggregator.
7. **Identity and KYC verification:** transmit ID scans and signatures as shares between branch, processor, and vault; tampered crops fail closed before approval.
8. **Journalism and source protection:** separate custody of sensitive photo evidence among editors and counsel; publication requires quorum consent encoded cryptographically rather than procedurally.

9. **Disaster-resilient records:** survive loss of minority shares to fire, flood, or seizure while preserving secrecy of survivors; recovery needs only any surviving quorum.
10. **Classroom and laboratory platform:** hands-on threshold, entropy, and integrity experiments with replayable sessions and comparable scorecards across student groups.

Deployment guidance: select (2, 2) for minimal two-party escrow, (2, 3) for single-fault tolerance, and (3, 5) for committee custody with dual-fault tolerance. Prefer XOR recovery for archival fidelity and OR stacking for on-screen or printed inspection. Always enable hash gating; never decode on partial or stale inputs.

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